Computer Networks
CL3001

LAB - 02
Introduction to Cisco Packet Tracer & basic networking devices

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES, KARACHI CAMPUS
FAST SCHOOL OF COMPUTING (AI & DS, CS, CY, SE)
SPRING 2026

Computer Networks Lab 02

Course: Computer Networks (CL3001) **Semester:** Spring 2026
Instructor: Mr. Sameer Faisal

Note:
- Maintain discipline during the lab.
- Listen and follow the instructions as they are given.
- Just raise hand if you have any problem.
- Completing all tasks of each lab is compulsory.
- Get your lab checked at the end of the session.

Lab Objective

- Introduction to Cisco Packet Tracer & basic networking devices.

Introduction to Packet Tracer

Packet Tracer – Creating a New Topology

What is Packet Tracer? Packet Tracer is a protocol simulator developed by Dennis Frezzo and his team at Cisco Systems. Packet Tracer (PT) is a powerful and dynamic tool that displays the various protocols used in networking, in either Real Time or Simulation mode. This includes layer 2 protocols such as Ethernet and PPP, layer 3 protocols such as IP, ICMP, and ARP, and layer 4 protocols such as TCP and UDP. Routing protocols can also be traced.

Purpose: The purpose of this lab is to become familiar with building topologies in Packet Tracer.

STEP – I:
Start packet tracer

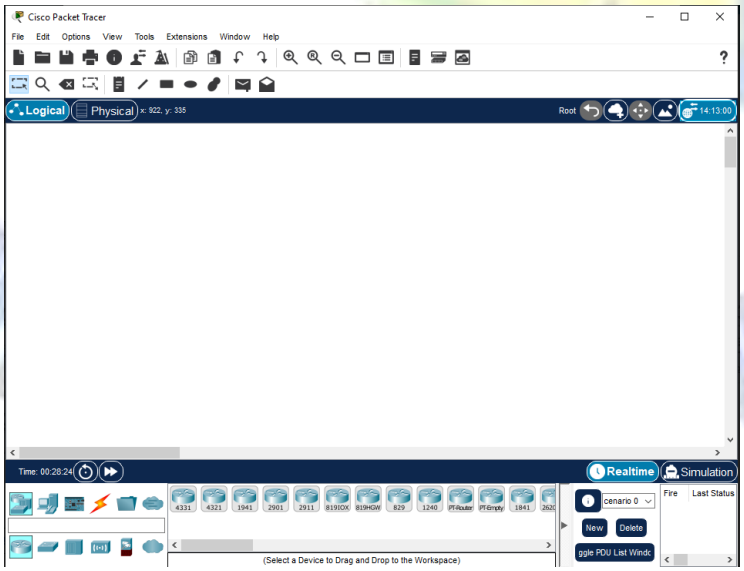


Figure 1 - Cisco Packet Tracer Interface

STEP – II:
Choosing Devices & Connections

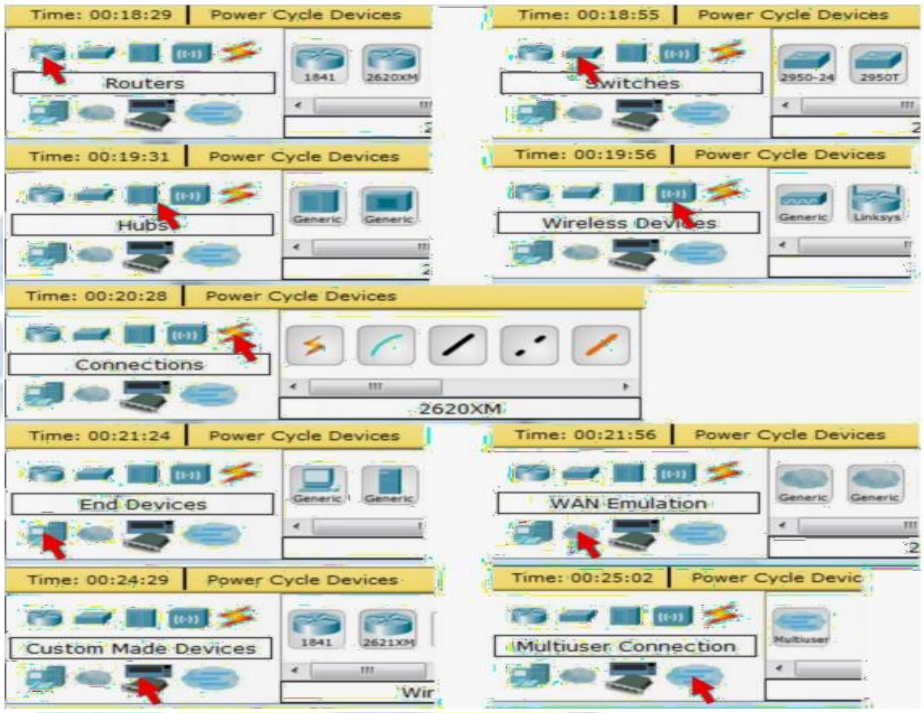


Figure 2 - Types of Devices in Cisco Packet Tracer

STEP – III:
Building the topology (adding the hosts)

Single click on the end devices.

Figure 3 - Adding Host Devices

Single click on generic host or any other required end device/host machine.

Figure 4 - Selecting a PC

Move the cursor into topology area. You will notice it turns into a ‘+’ sign.
Single click in the topology area and your selected end device gets copied in the topology area.

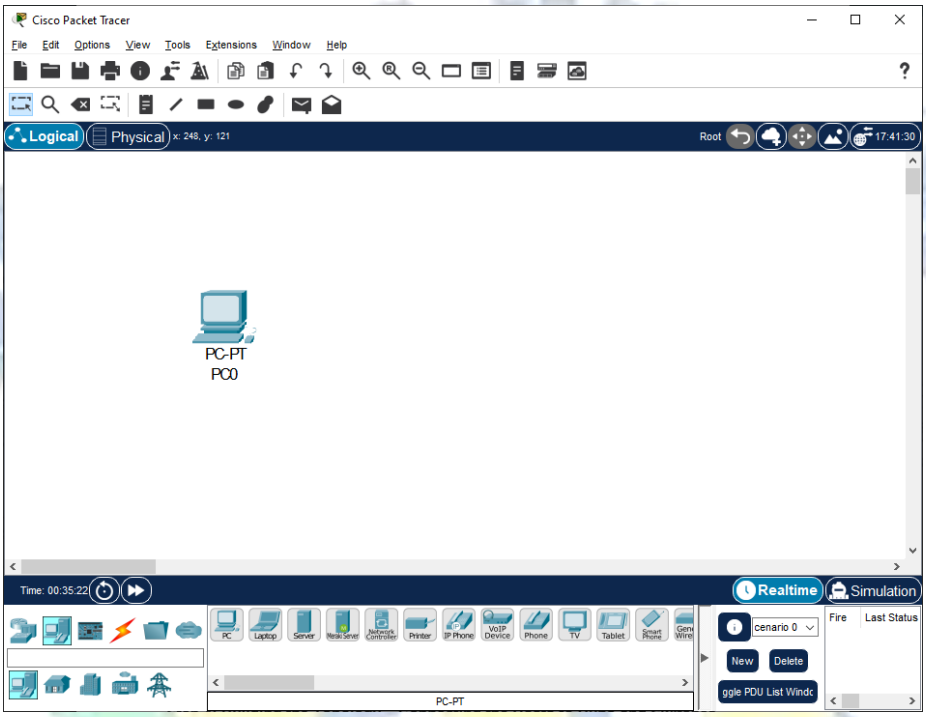


Figure 5 - Adding the devices to Cisco Packet Tracer

Let`s add 3 more hosts now (2 PCs and 1 Laptop).

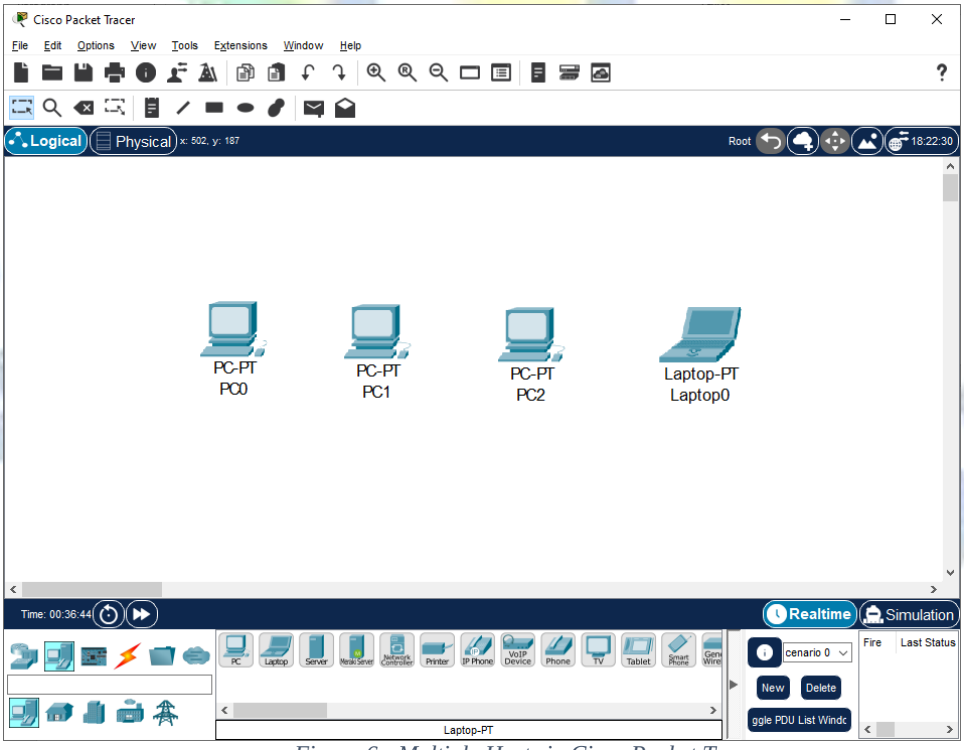


Figure 6 - Multiple Hosts in Cisco Packet Tracer

STEP – IV(a):
Building the topology (Connecting hosts to hosts).

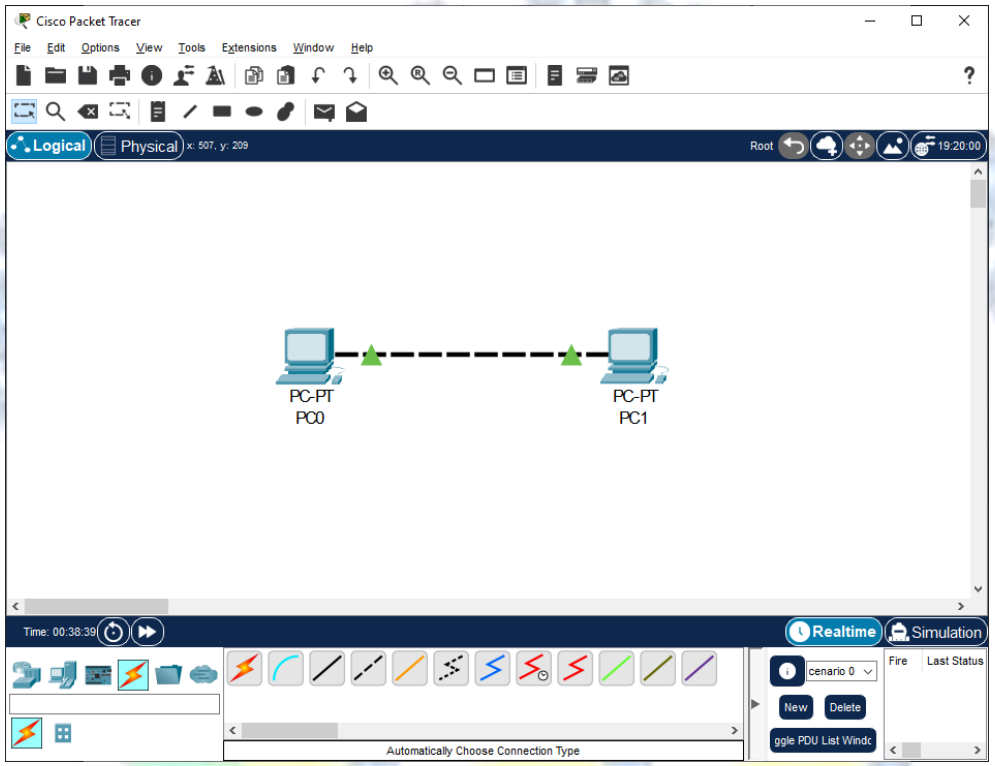


Figure 7 - Topology Building (Connecting host to host)

STEP – IV(b):
Building the topology (Connecting hosts to hubs and switches).

Adding a hub:
Select a hub, by clicking once on hubs and once on a generic hub.

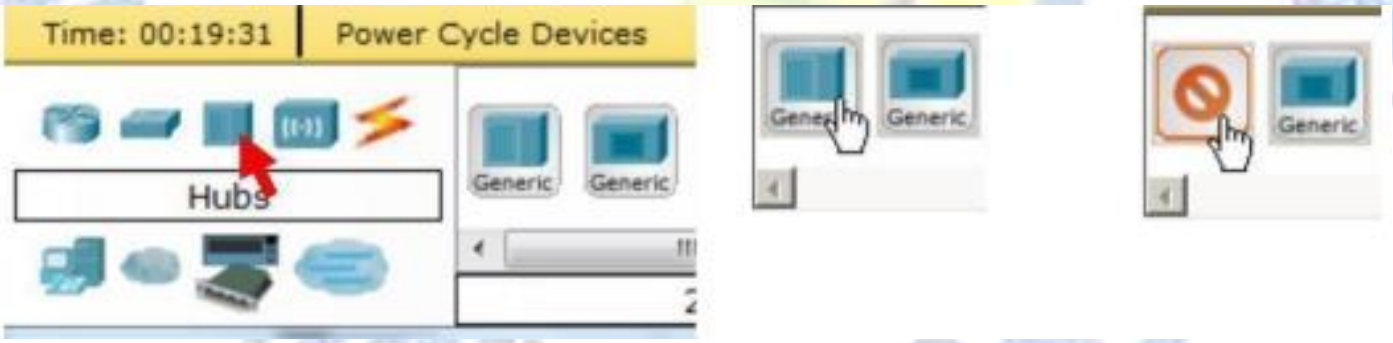
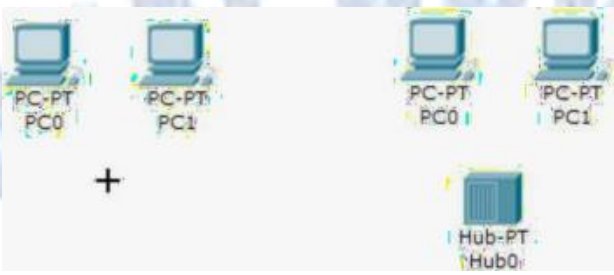


Figure 8 - Adding a Hub

Add the hub by moving the plus sign "+" below PC0 and PC1 and click once.



Connect PC0 to Hub0 by first choosing Connections.

Figure 9 - Wiring & Connections

Click once on the Copper Straight-through cable.

Figure 10 - Straight Through Cable

Perform the following steps to connect PC0 to Hub0:

1) Click once on PC0.
2) Choose FastEthernet.
3) Drag the cursor to Hub0.
4) Click once on Hub0 and choose Port0.
5) Notice the green link lights on both the PC0 Ethernet NIC and the Hub0 Port0 showing that the link is active.

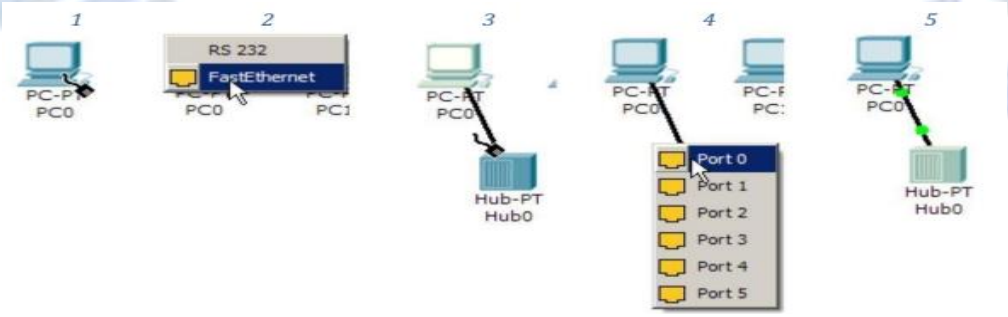


Figure 11 - Connecting Cables to Ports

Repeat the steps above for PC1 connecting it to Port l on Hub0. (The actual hub port you choose does not matter).

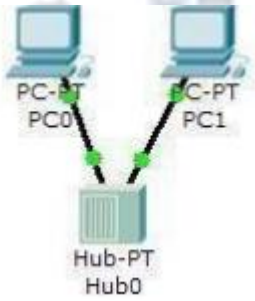


Figure 12 - PC connected to a Hub

Adding a switch.
Select a switch, by clicking once on Switches and once on a 2950-24 (2960 in newer versions) switch.

Figure 13 - Choosing the Switch

Add the switch by moving the plus sign '+' below PC2 and PC3 and click once.

Figure 14 - Adding a Switch

Connect PC2 to Switch0 by first choosing Connections.

Figure 15 - Wiring & Connections

Click once on the Copper Straight-through cable.

Figure 16 - Straight Through Cable

Perform the following steps to connect PC2 to Switch0:

1) Click once on PC2.
2) Choose FastEthernet.
3) Drag the cursor to Switch0.
4) Click once on Switch0 and choose FastEthernet0/1.
5) Notice the green link lights on PC2 Ethernet NIC and amber light Switch0 FastEthernet0/1 port. The switch port is temporarily not forwarding frames, while it goes through the stages for the Spanning Tree Protocol (STP) process.
6) After an about 30 seconds the amber light will change to green indicating that the port has entered the forwarding stage. Frames can now have forwarded out the switch port.

Note: Spanning Tree Protocol (STP) will be discussed later.

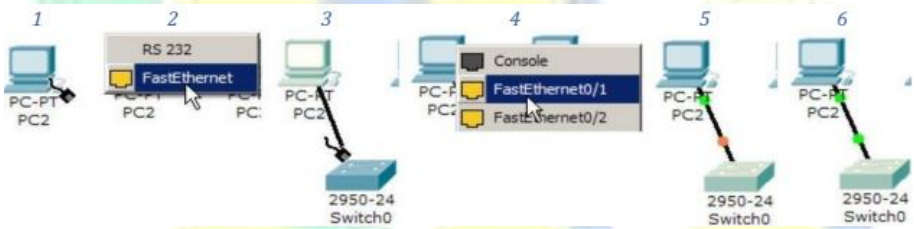


Figure 17 - Connecting PC to Switch

Repeat the steps above for PC3 connecting it to Port 3 on Switch0 on port FastEtherent0/2. (The actual switch port you choose does not matter.)

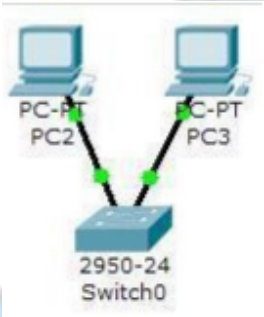


Figure 18 - Connecting PC to Switch

Move the cursor over the link light to view the port number. Fa means FastEthernet, 100 Mbps Ethernet.

STEP – V:

Configuring IP addresses and subnet masks on the hosts.

Before we can communicate between the hosts we need to configure IP Addresses and Subnet Mask on the devices.

1) Click once on PC0.

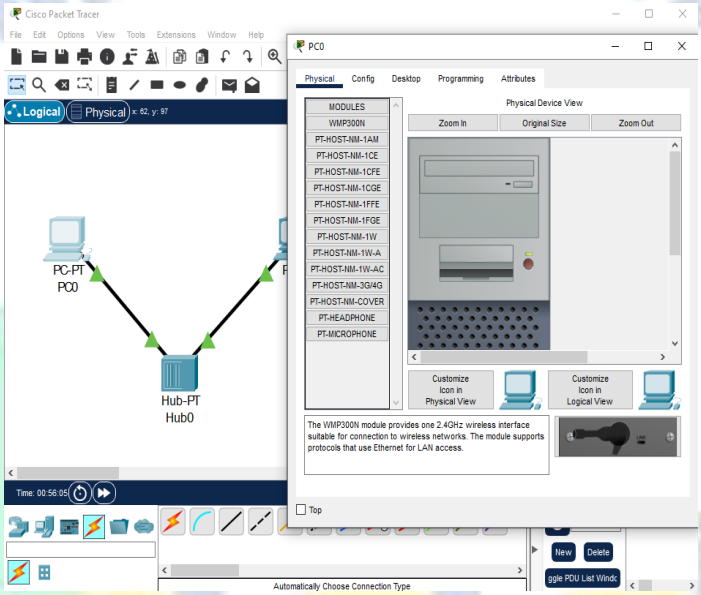


Figure 19 - Configuration Window in PC

2) Choose the Config tab and click on Settings. It is here that you can change the name of PC0.

3) Click on Interface and then FastEthernet. Although we have not yet discussed IP Addresses, add the IP Address to 172.16.1.10. Click once in the Subnet Mask field to enter the default Subnet Mask. You can leave this at 255.255.0.0. We will discuss this later.

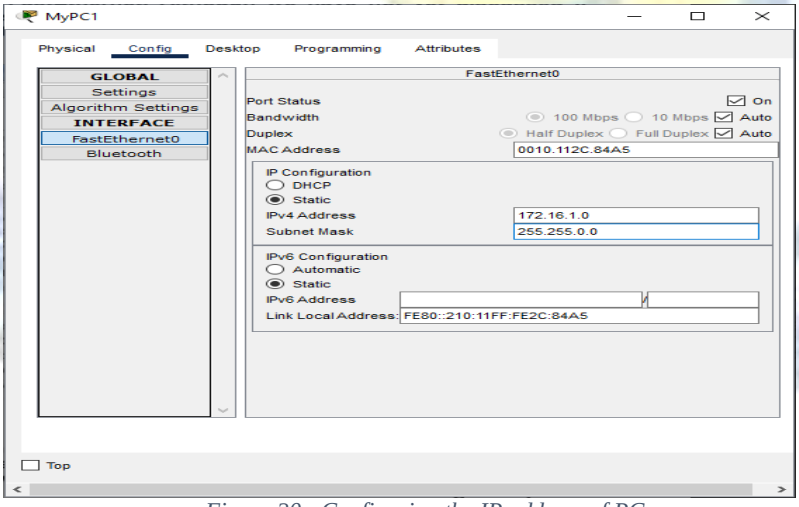


Figure 20 - Configuring the IP address of PC

Also, notice this is where you can change the Bandwidth (speed) and Duplex of the Ethernet NIC (Network Interface Card). The default is Auto (auto negotiation), which means the NIC will negotiate with the hub or switch. The bandwidth and/or duplex can be manually set by removing the check from the Auto box and choosing the specific option.

Bandwidth - Auto

If the host is connected to a hub or switch port which can do 100 Mbps, then the Ethernet NIC on the host will choose 100 Mbps (Fast Ethernet). Otherwise, if the hub or switch port can only do 10 Mbps, then the Ethernet NIC on the host will choose 10 Mbps (Ethernet).

Duplex - Auto

Hub: If the host is connected to a hub, then the Ethernet NIC on the host will choose Half Duplex.

Switch: If the host is connected to a switch, and the switch port is configured as Full Duplex (or Auto negotiation), then the Ethernet NIC on the host will choose Full Duplex.

If the switch port is configured as Half Duplex, then the Ethernet NIC on the host will choose Half Duplex. (Full Duplex is a much more efficient option.)
The information is automatically saved when entered. To close this dialog box, click the "X" in the upper right.

Repeat these steps for the other hosts. Use the information below for IP Addresses and Subnet Masks.

Host	IP Address	Subnet Mask
PC0	172.16.1.10	255.255.0.0
PC1	172.16.1.11	255.255.0.0
PC2	172.16.1.12	255.255.0.0
PC3	172.16.1.13	255.255.0.0

To verify the information that you entered, move the Select tool (arrow) over each host.

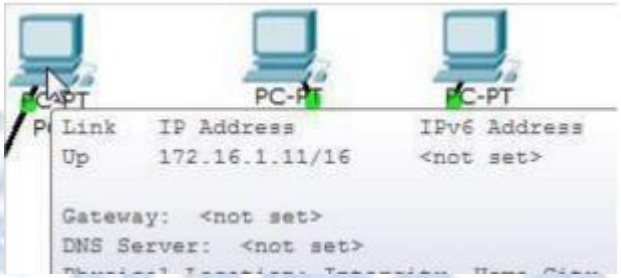


Figure 21 - Verifying the IP address

STEP – VI (Optional):
Deleting a device or link.
To delete a device or link, choose the Delete tool and click on the item you wish to delete.

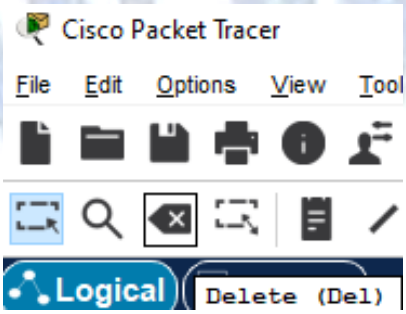


Figure 22 - Delete Tool

STEP – VII:
Connecting Hub0 to Switch0.
To connect like-devices, like a Hub and a Switch, we will use a Cross-over cable.
Click once the Crossover Cable from the Connections options.

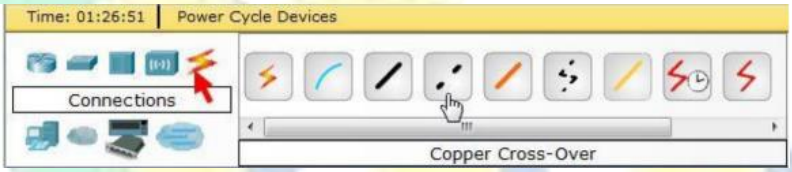


Figure 23 - Cross-Over Cable

Move the Connections cursor over Hub0 and click once.

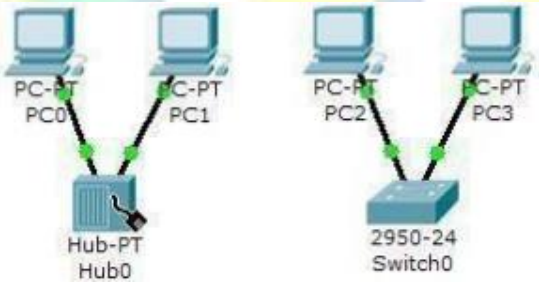


Figure 24 - Connections to Switch & Hub

Select Port 5 (actual port does not matter).

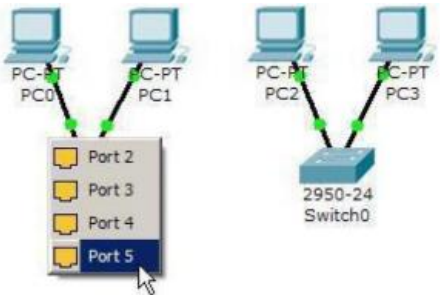


Figure 25 - Port Selection

Move the Connections cursor to Switch0.

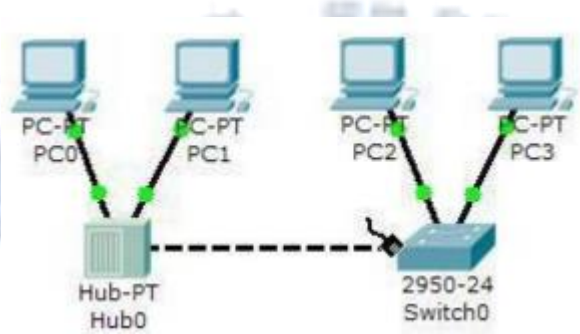


Figure 26 - Hub to Switch Connection

Click once on Switch0 and choose FastEthernet0/4 (actual port does not matter).

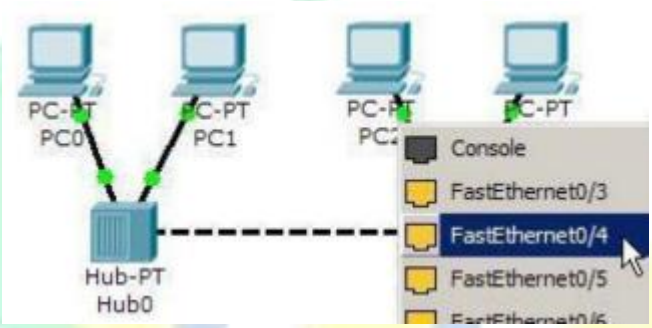


Figure 27 - - Hub to Switch Connection

The link light for switch port FastEthernet0/4 will begin as amber and eventually changes to green as the Spanning Tree Protocol transitions the port to forwarding.

1

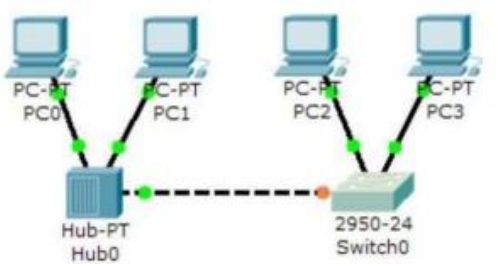


Figure 28 - Spanning Tree Protocol

2

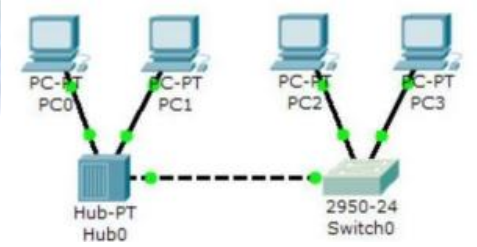


Figure 29 - Spanning Tree Protocol

STEP – VIII:
Verifying connectivity in ‘Realtime’ mode.

Be sure you are in Realtime mode.

Figure 30 - Realtime Mode

Select the Add Simple PDU tool used to ping devices.

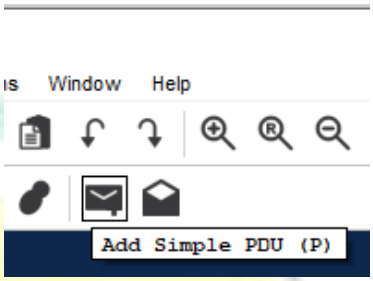


Figure 31 - Protocol Data Unit (PDU)

Click once on PC0, then once on PC3.

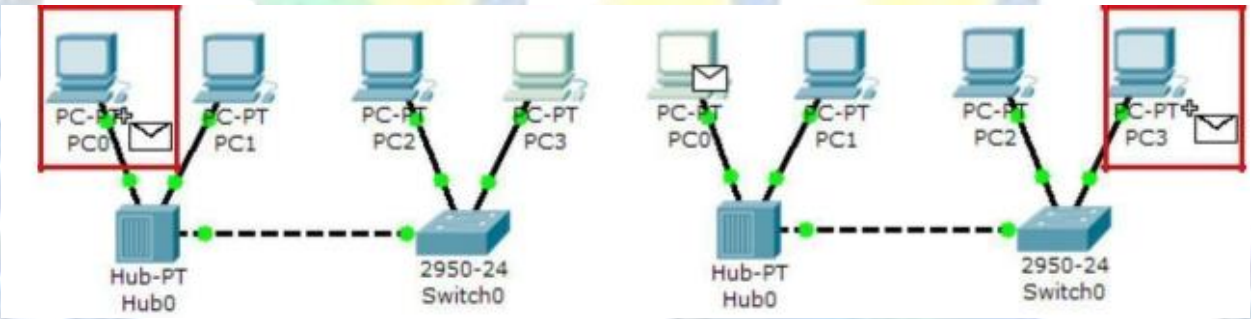


Figure 32 - Sending PDU from Source to Destination

The PDU Last Status should show as Successful.

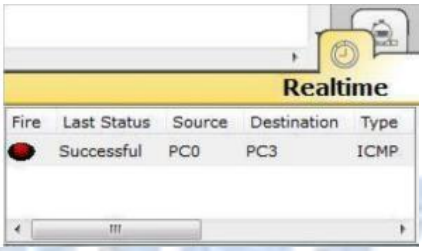


Figure 33 - PDU Verification

STEP – IX:
Resetting the network.
At this point we will want to reset the network, whenever you want to reset the network and begin the simulation again, perform the following tasks:

1) Click Delete in the PDU area.

Figure 34 - Delete PDU

2) Now, Power Cycle Devices and confirm the action.

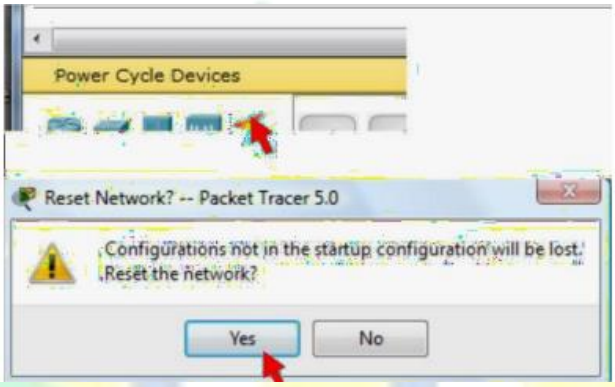


Figure 35 - Resetting the Network

3) Waiting for Spanning Tree Protocol(STP).

Note: Because Packet Tracer also simulates the Spanning Tree Protocol, at times the switch may show amber lights on its interfaces. You will need to wait for the lights to turn green on the switches before they will forward any Ethernet frames.

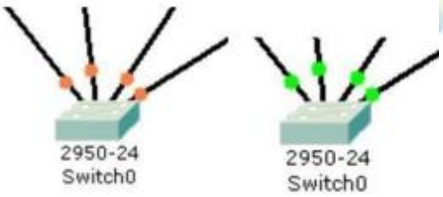


Figure 36 - STP

STEP – X (Optional):
Verifying Connectivity in Simulation Mode.
Be sure you are in Simulation mode.

Figure 37 - Simulation Mode

Deselect all filters (All/None) and select only ICMP.

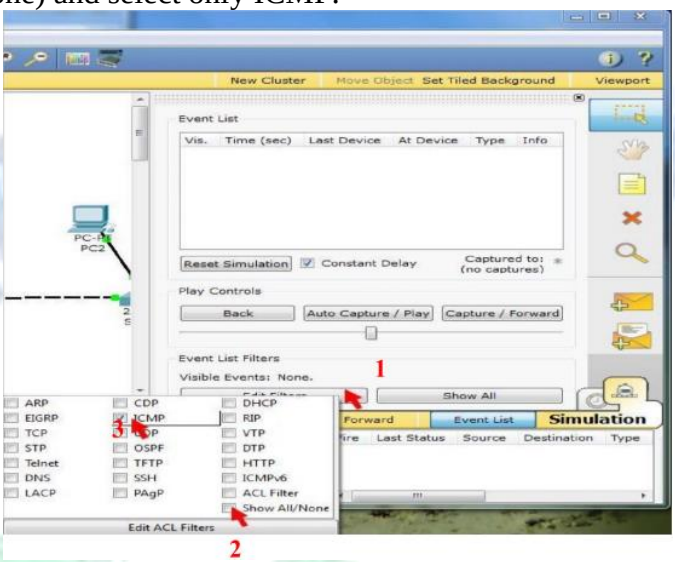


Figure 38 - Simulation of Protocols

Select the Add Simple PDU tool used to ping device.

Click once on PC0, then once on PC3.

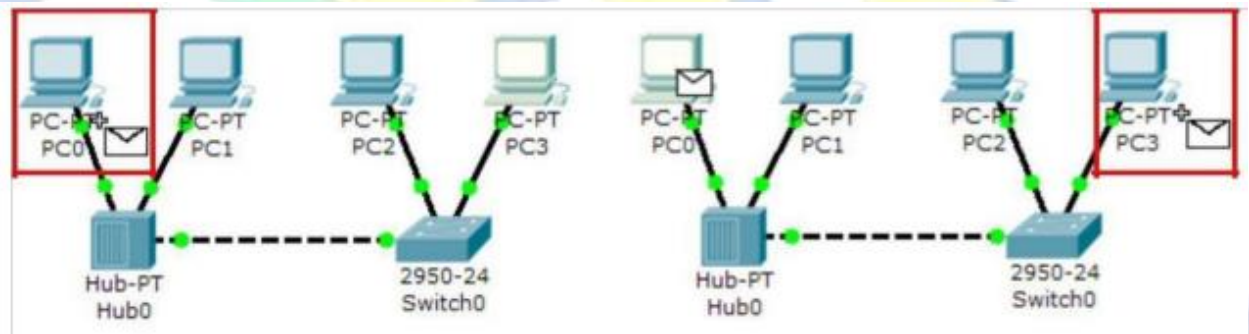


Figure 39 - Sending PDU from Source to Destination

Continue clicking Capture/Forward button until the ICMP ping is completed. You should see the ICMP messages move between the hosts, hub and switch. The PDU Last Status should show as Successful. Click on Clear Event List if you do not want to look at the events or click Preview Previous Events if you do. For this exercise it does not matter.

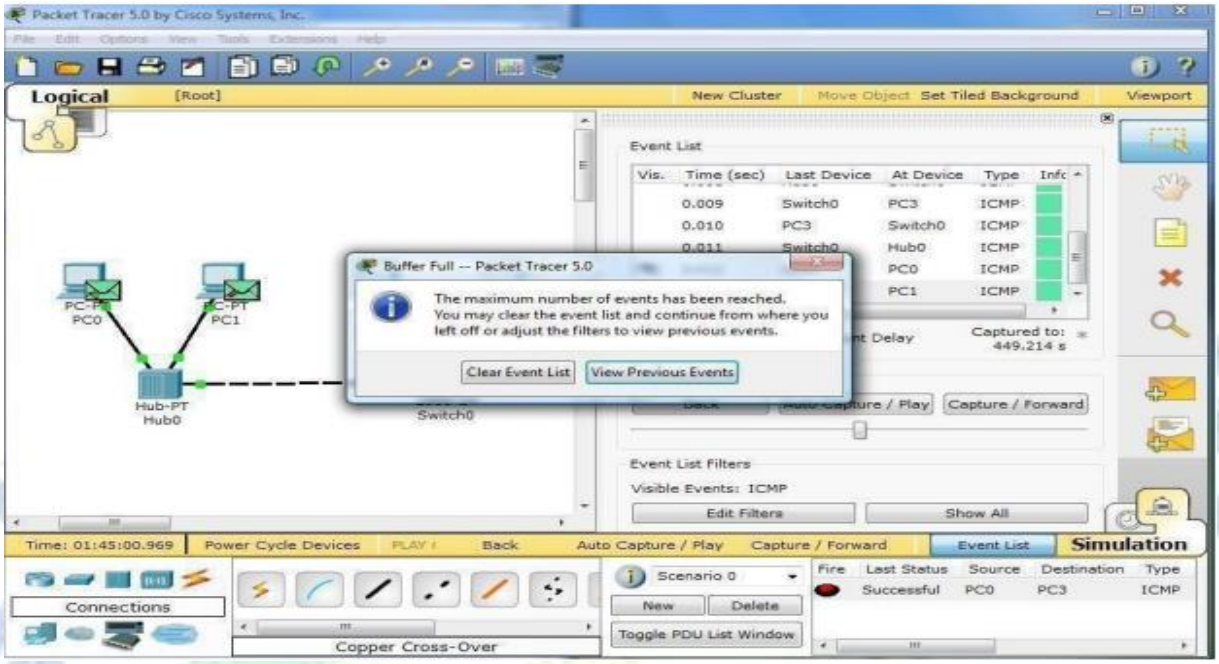


Figure 40 - Clearing Simulation Events

STEP – XI:

Saving the topology.

Perform the following steps to save the topology (uses .pkt file extension).

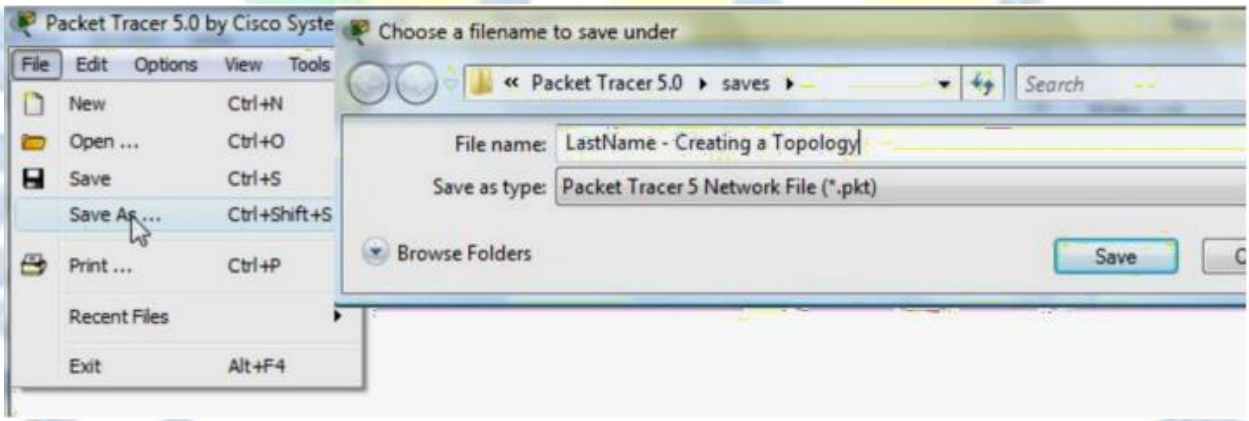


Figure 41 - Saving Your Work (Topology)

Opening Existing PT Topologies:

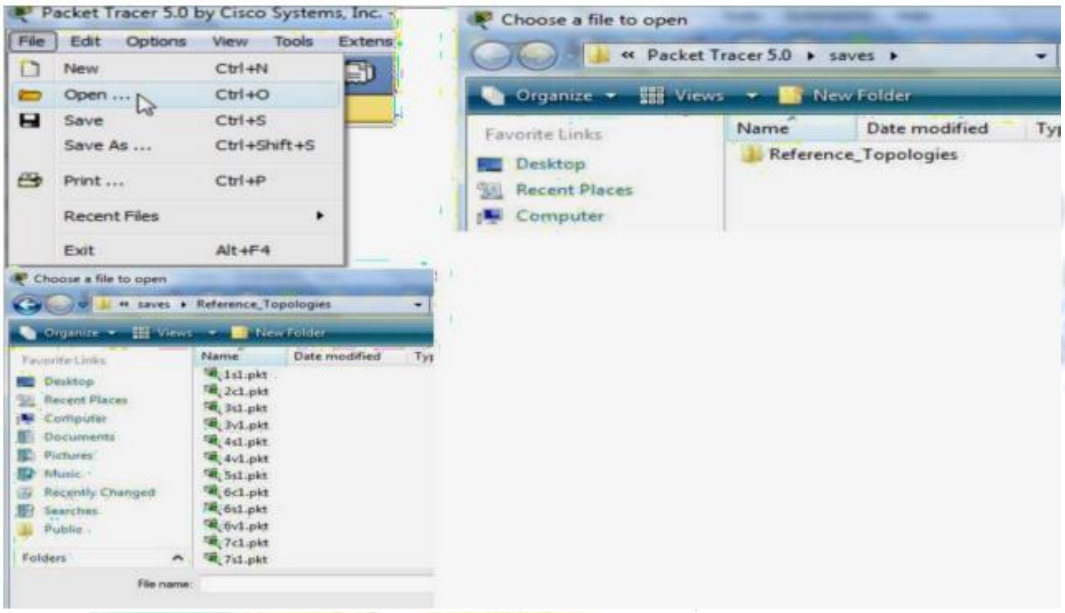


Figure 42 - Opening Existing Packet Tracer Topologies

DHCP

It stands for **dynamic host configuration protocol**; it is a network management protocol used mainly to distribute IP address to all connected devices in a network. Whenever a system is turned on in a network it sends a request to server this request is called **DHCP DISCOVER**.
On receiving the request, it supplies the client with a IP address from its pool this offer is called **DHCP OFFER**, on receiving the offer the client responds with the packet called **DHCP REQUEST** to eventually get the IP.

a) Select 3 PCs and a server from "End Devices" your work area will look like this:

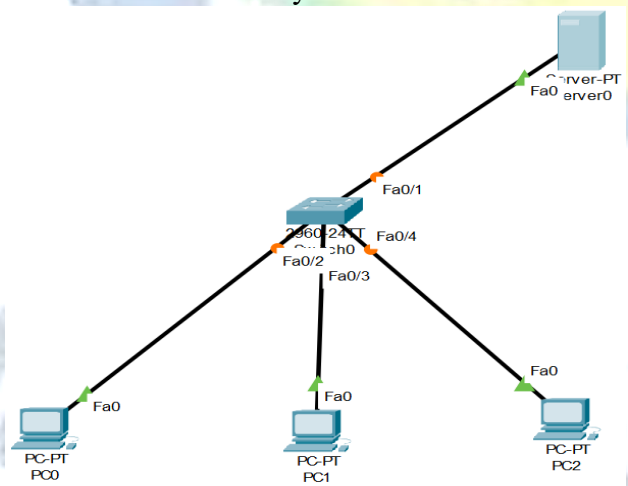


Figure 43 - Topology Creation

b) Choose “Connection Type” to connect these devices with Switch.
c) Now click the Server. The screen you will see will look something like this:

Figure 44 - Choose Services on Server

d) Here go to Fast Ethernet placed just under Interface marked with red. After clicking “Fast Ethernet” you will see following window. Here assign IP ADDRESS to this SERVER:

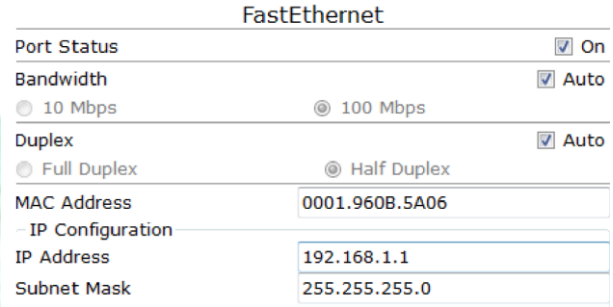


Figure 45 - Server IP Assignment

e) Now click “DHCP” to make this server a DHCP server:

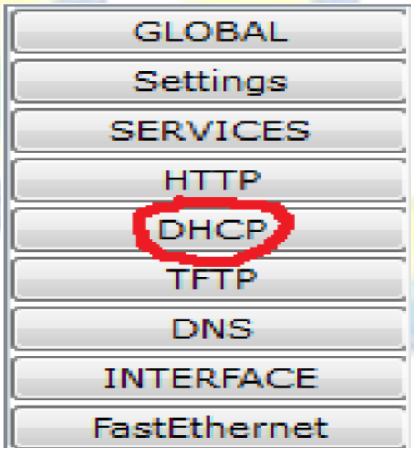


Figure 46 - Select DHCP Service on Server

f) DHCP is “Dynamic Host Control Protocol” a DHCP server is responsible for assigning IP Addresses to the devices present in the Network from its pool. This concept can be seen in the image shown below:

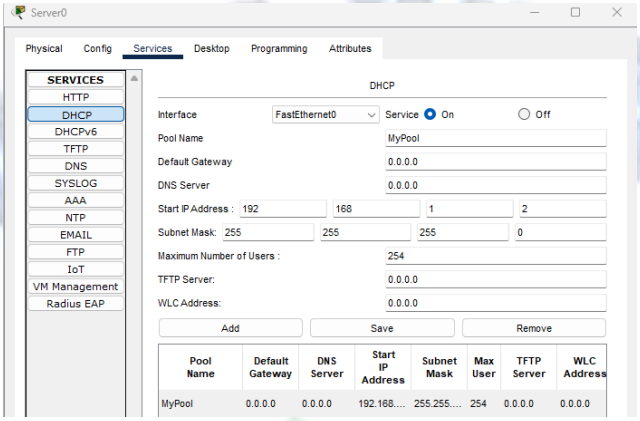


Figure 47 - Creating DHCP Pool

g) Now each device will take IP from DHCP. Now click on all three PCs, go to “DESKTOP”, then “IP CONFIGURATION”, here select option “DHCP” instead of assigning IP Address statically. One PC is shown below:

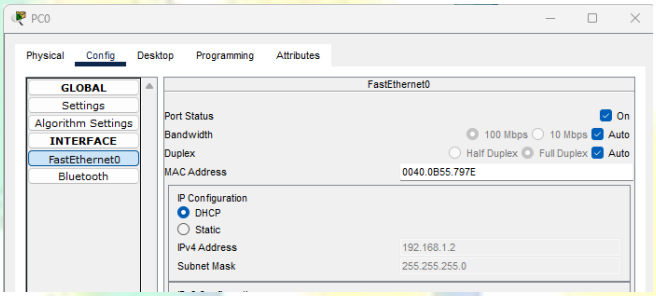


Figure 48 - Assignment of IP address to Hosts Through DHCP

Simulation of Different Topologies

1) Bus Topology

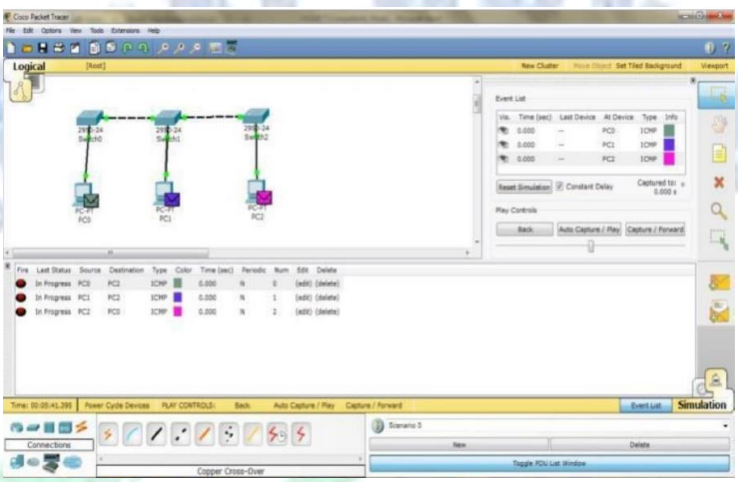


Figure 49 - Bus Topology

2) Star Topology

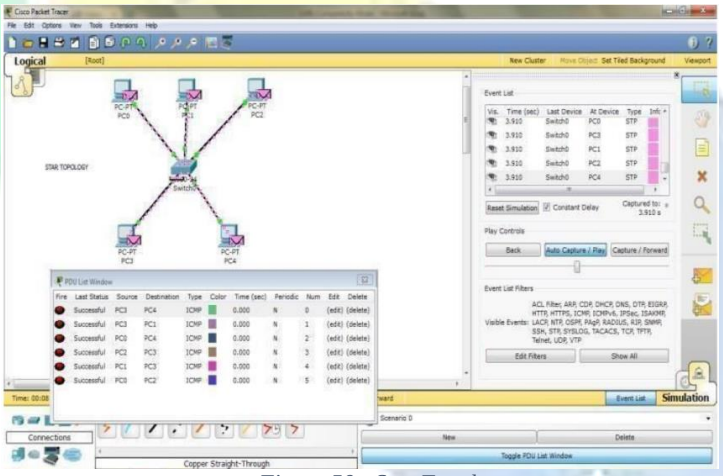


Figure 50 - Star Topology